Why people punish – Simulation

1. Setup

Individuals are endowed with a strategy for the repeated prisoner's dilemma, represented by a finite-state automaton (see Section 2). They are assigned to groups of size $N = 6$ and are randomly rematched for $T = 20$ supergames.

At the start of each match, players begin in state 1 and play an indefinitely repeated prisoner's dilemma with a punishment option. The continuation probability of the repeated game is $\delta = 0.7$.

The stage-game matrix is parametrized by:

- b and c: benefits and costs from cooperation;
- α and β: cost and damage of punishment.

The resulting payoff matrix is is:

	C	D	P
C	b-c	-c	-c- β
D	b	0	- β
P	b-α	-α	-α- β

To avoid negative payoffs, I apply a linear transformation with parameters l and m:

	C	D	P
C	l(b-c)+m	l(-c)+m	l(-c- β)+m
D	l(b)+m	m	l(- β)+m
P	l(b-α)+m	l(-α)	l(-α- β)+m

We consider two treatments:

	Non-cooperative treatment	Cooperative treatment
b	4	2
c	3	1
α	3	1
β	12	4
l	1	3

m	15	15

So that the resulting payoff matrices are, for the cooperative treatment:

	C	D	P
C	18	12	0
D	21	15	3
P	18	12	0

And for the non-cooperative treatment:

	C	D	P
C	16	12	0
D	19	15	3
P	16	12	0

2. Strategies

Each subject is characterized by a strategy. Strategies determine behavior within a supergame.

We focus on three types:

1. non-punishing cooperator (grim-trigger);
2. punishing cooperator;
3. defector.

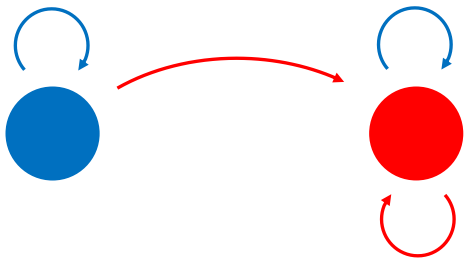
The initial distribution of strategies in the group varies across sessions and is given by: $(N_c,\ N_P,\ N_D)$, representing respectively the number of non-punishing cooperators, punishing cooperators and defectors.

Strategies are represented by finite-state automata. Each auotomaton is represented (like in Chris' simulations) like a matrix where:

- Each column represent a state;
- The first row encodes the action in each state;
- Each of the following rows encodes the destination state after a certain pair of actions has been observed.

	STATE 1	STATE 2	STATE 3	...
ACTION	Action in state 1	Action in state 2	Action in state 3	
CC	If in state 1 and CC observed, which state to transition to	If in state 2 and CC observed, which state to transition to	...	
CD	If in state 1 and CD observed, which state to transition to	...		
CP	...			
DC				
DD				
DP				
PC				
PD				
PP				

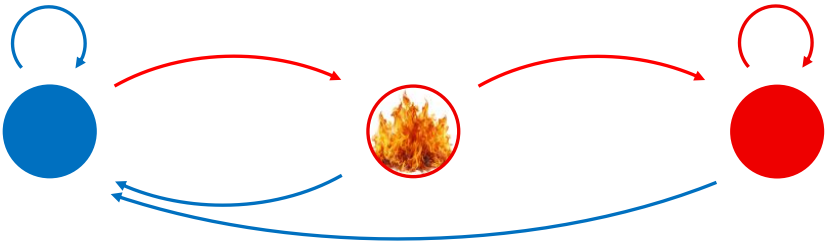
2.1 Non-punishing cooperator (Grim-Trigger)



	STATE 1	STATE 2
ACTION	C	D
CC	1	
CD	2	
CP	2	
DC		2
DD		2

DP		2
PC		
PD		
PP		

2.2 Punishing cooperators



	STATE 1	STATE 2	STATE 3
ACTION	C	P	D
CC	1		
CD	2		
CP	2		
DC			1
DD			3
DP			3
PC		1	
PD		3	
PP		3	

2.3 Defector

	STATE 1
ACTION	D
CC	
CD	

CP	
DC	1
DD	1
DP	1
PC	
PD	
PP	

3. Meta-strategies

At the end of each supergame, one randomly selected individual in the may revise their strategy.

The update rule is as follows:

- The individual evaluates all supergames since their last update
- They compute their average per-round payoff
- They compare it with counterfactual payoffs from the two alternative strategies[1]
- If one alternative yields a strictly higher payoff, they switch to that strategy
- If both alternatives yield equal payoffs, the choice is random

Additionally, punishers receive a payoff adjustment; their average per-round payoff is increased by a term "punisher_bias" (default set at 1).

4. Data generation

I generate data for 12 sessions. In each session there are 4 matching groups. Each group is randomly composed, and in particular:

$$N_c \sim U[1,3]$$
$$N_N \sim U[1,3]$$
$$N_D = N - N_C - N_N$$

Within each session, two groups are assigned to the cooperative treatment, and two to the non-cooperative treatment.

The resulting simulated dataset has the following columns:

Column	Explanation
session_id	Identifier for the session
group_id	Identifier for the group
treatment	Either "cooperative" or "non-cooperative", with two groups per treatment per session

[1] The counterfactual payoff is computed by holding the length of each supergame equal to its realized value, while allowing action profiles to differ. We assume that the agent knows the types of her opponents and recomputes the payoff she would have obtained under an alternative strategy, given opponents' strategies and the realized duration of each round.

Nc	Number of non-punishing cooperators at the beginning of a session
Np	Number of punishing cooperators at the beginning of a session
Nd	Number of defectors at the beginning of a session
supergame	Number of the supergame
pair_id	Identifier of the current pair
round	Current round in the supergame
player_id	Identifier of the player in the group
partner_id	Identifier of the opponent in the group
type_start	Player type at the beginning of the session
type_current	Current player type
action	Action played in that round (either C,D or P)
opponent_action	Opponent's action in that round
payoff	Payoff for that round
cum_payoff	Cumulative payoff

5. Analysis

Work in progress...